IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies.**

1. Collect and integrate data using SpaceX API and web scraping.Clean, structure, and prepare data for analysis.
2. Perform exploratory analysis through SQL queries and data visualization.
3. Build interactive tools with Folium maps and Plotly Dash.
4. Train, tune, and evaluate classification models.

- **Summary of all results.**

1. Identify key predictors of landing success: launch site, orbit type, and payload mass.
2. Generate interactive visualizations and an analytics dashboard.
3. Develop a predictive model achieving 83.33 % classification accuracy.
4. Provide data-driven insights to inform launch cost and risk assessment.

GitHub Repository.

https://github.com/gogetafterit/spacex_falcon9_analysis.git

Introduction

• **Project background and context.**

SpaceX can launch a Falcon 9 for tens of millions less than its competitors. This cost advantage depends on recovering and reusing the first-stage booster. Accurately predicting a successful booster landing is therefore fundamental for estimating launch costs and assessing mission risk. This project uses data science to analyze what makes a landing succeed.

• **Problems you want to find answers.**

What are the main factors that determine if a Falcon 9 first stage will land successfully?

Which launch sites and mission profiles have the highest success rates?

Can we build a reliable machine learning model to predict landing outcomes?

How can data-driven insights inform cost estimation and competitive strategy in the launch industry?

Section 1

Methodology

Methodology (1/3).

Executive Summary

- **Data collection methodology:**

Data was collected using the SpaceX API and web scraping.

- **Perform data wrangling.**

The raw data was cleaned, standardized, and restructured for analysis.

- **Perform exploratory data analysis (EDA) using visualization and SQL.**

SQL queries and data visualizations were used to explore patterns and relationships.

- **Perform interactive visual analytics using Folium and Plotly Dash.**

An interactive Folium map and a Plotly Dash dashboard were developed.

- **Perform predictive analysis using classification models.**

Classification models were built, tuned, and evaluated to predict landing success.

Methodology (2/3).

- Data Collection:

Retrieved data via SpaceX REST API and web scraping.

- Data Wrangling:

Cleaned and standardized raw data using Pandas (.fillna(), .value_counts()); engineered landing outcome labels.

- Exploratory Data Analysis (EDA):

Used SQL and Matplotlib to identify launch patterns and relationships.

Methodology (3/3).

- Interactive Visual Analytics:

Developed a geospatial Folium map and an interactive Plotly Dash dashboard.

- Predictive Modeling:

Trained classification models (Logistic Regression, SVM, KNN, Decision Trees) using Scikit-Learn.

- Evaluation:

Optimized hyperparameters with GridSearchCV and assessed performance via confusion matrices and accuracy.

Data Collection (Overview).

- **Sources**: SpaceX REST API, Wikipedia (Web Scraping)
- **Methods**: HTTP GET, JSON parsing & normalization, HTML extraction (BeautifulSoup)
- **Output**: Two CSV datasets

SpaceX API -> GET request -> JSON -> DataFrame -> Export CSVand thenWikipedia -> HTML -> BeautifulSoup -> Parse tables -> Export CSV

Data Collection – SpaceX API

- **Source**: SpaceX REST API (api.spacexdata.com/v4/launches/past)
- **Process**: GET JSON -> pd.json_normalize() -> DataFrame -> extract features (Booster, Launch site, Payload, Core)
- **Preparation**: Filter Falcon 9 -> fill missing values (mean)
- **Output**: dataset_part_1.csv
- **GitHub**: https://github.com/gogetafterit/spacex_falcon9_analysis.git

SpaceX API -> HTTP GET request -> JSON response -> pd.json_normalize() -> DataFrame -> Apply custom functions to extract features -> Filter for Falcon 9 -> Clean data -> Export CSV

Data Collection - Scraping

- **Source**: Wikipedia (June 2021)
- **Process**: GET -> HTML -> BeautifulSoup -> parse tables -> extract headers & rows -> clean (date_time(), booster_version(), get_mass(), landing_status())
- **Output**: DataFrame -> CSV
- **Tool**: BeautifulSoup4
- **GitHub**: https://github.com/gogetafterit/spacex_falcon9_analysis.git

Wikipedia URL -> HTTP GET -> HTML response -> BeautifulSoup object -> Locate tables (find_all) -> Extract headers -> Parse rows -> Clean data (date_time, booster_version, get_mass, landing_status) -> Build dictionary -> Convert to DataFrame -> Export CSV

Data Wrangling.

Load: dataset_part_1.csv

Columns: numeric & categorical

Check: missing values (%)

EDA: launches/site, orbit types, mission outcomes

Class: 1 = success, 0 = failure

Clean: fill NaN

Export: dataset_part_2.csv

GitHub: https://github.com/gogetafterit/spacex_falcon9_analysis.git

Load dataset_part_1.csv -> Identify columns (numeric & categorical) -> Check for missing values -> Exploratory analysis -> Create binary class -> Clean data -> Export dataset_part_2.csv

EDA with Data Visualization.

Findings

- CCAFS SLC-40 has the most launches
- KSC LC-39A shows higher success rates
- Success increases with payload
- Some orbits achieved 100% success
- 2013–2017: success trend rising

Feature Engineering

- One-hot encoding applied

Export: dataset_part_3.csv

Tools: Pandas, Matplotlib, Seaborn

GitHub:

Load dataset_part_2.csv -> EDA visualizations: [FlightNum vs LaunchSite | Payload vs LaunchSite | SuccessRate by Orbit | FlightNum vs Orbit | Payload vs Orbit | Yearly Trend] -> Feature engineering: [Select features -> One-Hot Encoding -> Convert to float64] -> Export dataset_part_3.csv -> Upload to GitHub

EDA with SQL.

SQL Queries Executed

- • Unique launch sites – `DISTINCT`
  • Filter sites 'CCA%' – `LIKE, LIMIT 5`
  • Total NASA payload – `SUM()`
  • Avg payload F9 v1.1 – `AVG()`
  • First ground success – `MIN(Date)`
  • Boosters (drone ship, 4000–6000 kg)
  • Mission outcomes – `GROUP BY`
  • Max payload boosters – Subquery
  • 2015 failures – Date filtering
  • Outcome rankings 2010–2017 – `ORDER BY DESC`
- **Key Functions Used**
  DISTINCT, LIKE, SUM, AVG, MIN, MAX, GROUP BY, Subqueries, DATE functions

Connect -> Load CSV -> Run SQL queries (DISTINCT, LIKE, SUM, AVG, MIN, MAX, GROUP BY, Subquery, ORDER BY) -> Extract results -> Analyze insights -> Publish GitHub URL

Build an Interactive Map with Folium.

Objects

- Map centered on NASA JSC
- Circles & markers for 4 launch sites
- MarkerCluster: 100+ launches (Green = success, Red = failure)
- Lines show distances; MousePosition shows live coordinates

Key Distances (CCAFS LC-40)

- Coast ~0.87 km | Railway ~1.29 km | Highway ~0.59 km | City ~51.7 km

Tools: Folium, Pandas, Haversine

GitHub: https://github.com/gogetafterit/spacex_falcon9_analysis.git

Load the Geo CSV -> Create a Folium map centered on the NASA center -> TASK 1: Add circles and markers for the four launch sites -> TASK 2: Add a MarkerCluster with color-coded launches (Green = Success, Red = Failure) -> TASK 3: Add MousePosition, calculate distances (Haversine), and draw polylines to the coastline, railway, highway, and city -> Save the map -> Export as HTML file -> Upload to GitHub

Build a Dashboard with Plotly Dash.

Dropdown: Choose launch site

- Pie Chart: Distribution of launch success
- Range Slider: Payload (0–10,000 kg)
- Scatter Plot: Payload vs outcome (booster shown by color)

Insights

- KSC LC-39A shows the highest success rate
- Payloads between 0–4000 kg succeed more often
- Newer boosters (B5) show higher success

GitHub: https://github.com/gogetafterit/spacex_falcon9_analysis.git

Load the dataset -> Select a launch site from the dropdown -> Pie chart updates (callback): All sites show success by site | Single site shows success vs failure -> Set the payload slider: 0–10,000 kg -> Scatter plot updates (callback): filtered by site and payload | Booster type shown by color -> Interactive dashboard -> Generate insights -> Publish on GitHub

Build a Dashboard with Plotly Dash.

Components: Dropdown (All/specific), Pie Chart (success distribution), Range Slider (0–10,000 kg), Scatter Plot (Payload vs outcome, color = booster)

Insights: KSC LC-39A highest success (76.9%), 0–4000 kg payloads succeed more, new boosters (B5) higher success

Tools: Dash, Plotly, Pandas, google.colab.output

GitHub: https://github.com/gogetafterit/spacex_falcon9_analysis.git

Load spacex_launch_dash.csv -> Create Dash app (Dropdown + Charts) -> Dropdown: Launch Site (ALL/Specific) -> Pie Chart (callback): Success distribution -> Range Slider: Payload 0–10,000 kg (step 1000) -> Scatter Plot (callback): Payload vs Outcome -> Run on proxyPort 8050 -> Public URL -> Upload to GitHub

Predictive Analysis (Classification).

Predictive AnalysisData: 80/20 split; features standardized

Models: Logistic Regression, SVM, Decision Tree, KNNTuning: 10-fold cross-validation

Results: All models reached 83.33% test accuracy

Best Model: Decision Tree (highest training score, clear logic)

1. Confusion Matrix[Insert heatmap screenshot]Note: Strong performance in predicting successful landings

2. ML WorkflowPrepare (standardize & split) -> Train (tune 4 algorithms) -> Test (accuracy + confusion matrix) -> Select best model: Decision Tree (83.33%)

Results.

EDA: Success increased after 2013 (>90%). KSC LC-39A highest (~76.9%). ES-L1, GEO, HEO, SSO show 100% success. Payload >7000 kg and more flight experience -> higher success.

Interactive Analytics: Folium map (launch sites + proximity). Dash dashboard: pie chart (success by site), scatter plot (payload vs outcome), filters (site + payload).

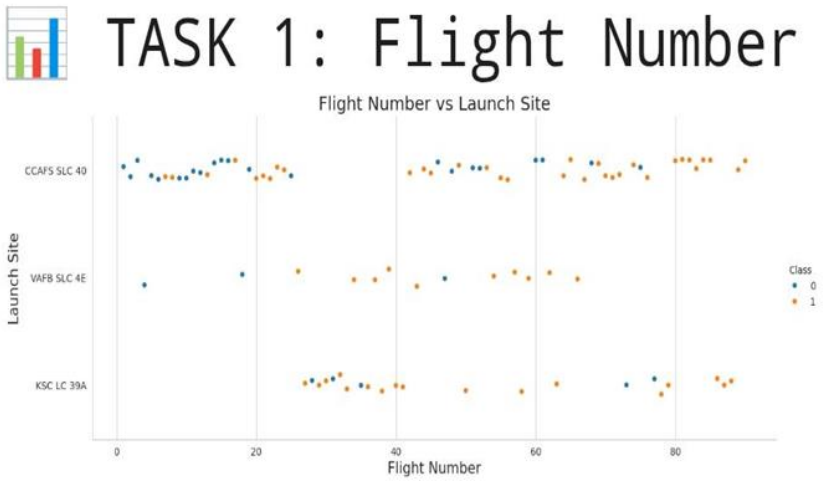
Predictive Analysis: Logistic Regression, SVM, Decision Tree, KNN. Best model: Decision Tree (~83–94%). Metric: confusion matrix. Impact: landing prediction -> cost advantage ($62M vs $165M).

Section 2

Insights drawn from EDA

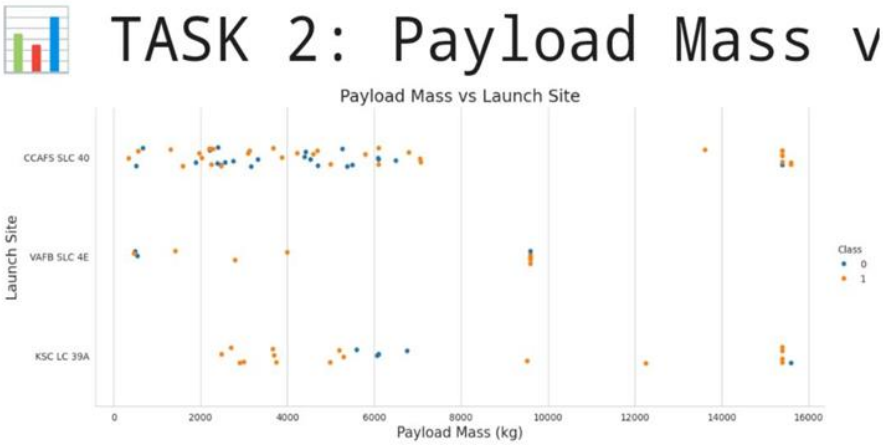
Flight Number vs. Launch Site.

As SpaceX gained experience (higher flight numbers), the probability of successful first-stage landing clearly increased, especially visible at CCAFS SLC 40, while the newer site KSC LC 39A started with already excellent success rates. This reflects SpaceX's rapid learning curve in booster recovery.



Payload vs. Launch Site.

Heavy payloads (>10,000–12,000 kg) are almost exclusively launched from CCAFS SLC 40 and tend to have a very high success rate for booster landing, while VAFB SLC 4E is used only for lighter payloads and shows more variable outcomes. This reflects different mission profiles and vehicle configurations used at each launch site.



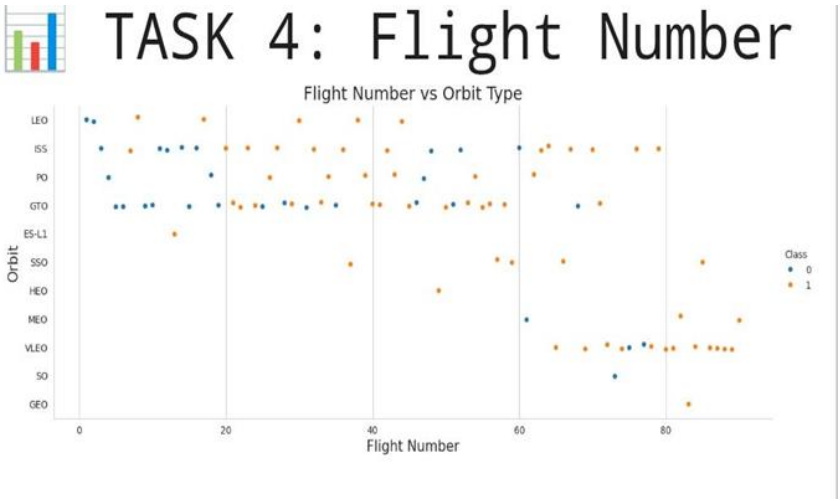
Success Rate vs. Orbit Type.

Orbits that are closer to Earth and/or have more forgiving re-entry profiles (especially ES-L1, GEO, SSO, VLEO) show significantly higher booster landing success rates, while GTO missions — which require much higher energy and more demanding re-entry conditions — have the lowest success rate (~50%). This is why predicting landing success is especially challenging (and valuable) for GTO missions.

Flight Number vs. Orbit Type.

The more flights SpaceX completed, the better they became at landing in LEO/ISS/VLEO/SSO orbits — but GTO missions remain significantly more difficult and show much lower and more inconsistent success rates, even on recent flights.

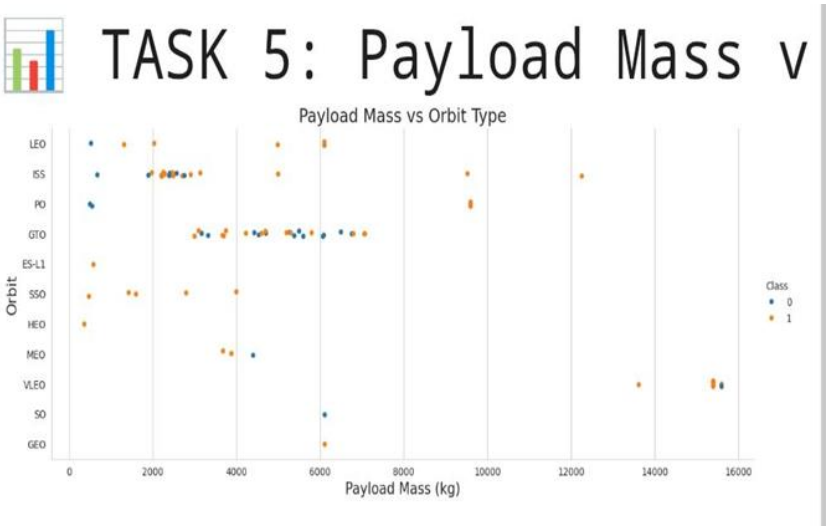
This is why GTO is often considered the real “final exam” for booster reusability.



Payload vs. Orbit Type.

Heavy payloads (>10,000–12,000 kg) are almost exclusively sent to LEO/ISS and show a very high landing success rate, while GTO missions — typically in the 3,000–6,500 kg range — remain much more difficult with roughly 50/50 success regardless of exact payload mass.

This again highlights why GTO is the most challenging orbit for reusable booster recovery.



Launch Success Yearly Trend.

SpaceX dramatically improved first-stage landing success — from basically 0% before 2015 to consistently 80–90% by 2019–2020 — showing extremely fast learning and technology maturation in just ~5 years.

This is the main reason reusable boosters became economically viable.

All Launch Site Names.

Unique Launch Sites

* sqlite:///spacex_database.db

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Note: LC-40 and SLC-40 refer to the same launch complex at Cape Canaveral; the difference comes from historical naming conventions in the dataset.

The dataset contains four real launch sites:

CCAFS LC-40 – Cape Canaveral Air Force Station, Launch Complex 40 (Florida).

VAFB SLC-4E – Vandenberg Air Force Base, Space Launch Complex 4E (California).

KSC LC-39A – Kennedy Space Center, Launch Complex 39A (Florida).

CCAFS SLC-40 – Cape Canaveral Air Force Station, Space Launch Complex 40 (Florida).

Launch Site Names Begin with 'CCA'.

Dataset: SpaceX Falcon 9 launches (2010–2013)

Goal: Predict landing success -> estimate launch cost

Method: API + web scraping; EDA, SQL, dashboards; ML (Logistic, SVM, Tree, KNN)

Results: Success rising after 2013; KSC LC-39A highest (~76.9%); Decision Tree best (~94.4%)

Impact: Cost estimate ($62M vs $165M+) for competitive bioscience p

Date	Time_UTC	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass.

NASA CRS Total Payload
Total_Payload_KG
45596

NASA CRS Total Payload: 45,596 kg — Total cargo carried by SpaceX Falcon 9 for NASA CRS (NASA Commercial Resupply Services) missions in the dataset, calculated with a SQL SUM() on Customer = 'NASA (CRS)'.

Average Payload Mass by F9 v1.1

Task: Average Payload for F9 v1.1

SQL Query:

```
SELECT AVG(PAYLOAD_MASS__KG_) AS
Avg_Payload_KG

FROM SPACEXTABLE

WHERE Booster_Version = 'F9 v1.1’;
```

Result:

- 2928.4 kg

Avg Payload: 2,928.4 kg — Average mass per Falcon 9 launch (~2.9 tons).

Shows typical mission scale and helps spot trends across launches.

Heavier payloads may link to specific orbits, launch sites, or booster versions.

Useful for EDA, feature engineering, and comparing payloads for success prediction.

First Successful Ground Landing Date.

Task 5: First Ground Pad Success

SQL Query:

```
SELECT MIN(Date) AS First_Success_Date
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

Result:

- 2015-12-22

This date marks SpaceX's first successful ground pad landing (Orbcomm OG2 Mission 2, Falcon 9 FT B1019 at CCAFS LC-40). This was a major milestone proving reusable rocket technology, enabling significant cost reductions for future launches.

Successful Drone Ship Landing with Payload between 4000 and 6000 kg.

Task 6: Successful Drone Ship Landings (4000–6000 kg)

SQL Query:

```
SELECT Booster_Version

FROM SPACEXTABLE

WHERE Landing_Outcome = 'Success (drone ship)'  AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Results:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Objective: Find Falcon 9 boosters that successfully landed on a drone ship with 4,000–6,000 kg payloads.

Method: SQL query filtering by Landing Outcome = Success (drone ship) and payload range 4000–6000 kg.

Result: 4 boosters found — all Falcon 9 Full Thrust (FT) variants.

Insight: FT boosters reliably handled medium-heavy payloads and drone ship landings, including reused boosters.

Total Number of Successful and Failure Mission Outcomes.

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Key Points:

Total Missions: 101.

Success: 100 (~99%).

Failures: 1 in-flight failure.

Note: 1 mission with unclear payload status.

Summary: Falcon 9 shows ~99% reliability across 101 missions, with only one recorded failure.

Boosters Carried Maximum Payload.

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

This table shows Falcon 9 Block 5 launches with payload masses. All missions carry 15,600 kg, likely Starlink batches, and use reused boosters, highlighting SpaceX's reusability and standardized launch strategy.

2015 Launch Records.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

The table shows two failed Falcon 9 v1.1 drone ship landing attempts from CCAFS LC-40 (boosters B1012 and B1015) in Jan and Apr 2015, reflecting early challenges in SpaceX's booster recovery efforts.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Total Launches: 31

Key Findings:

No landing attempt: 10 (32%)

Successful landings: 8 (5 drone ship, 3 ground pad)

Failed landings: 8 (mostly drone ship)

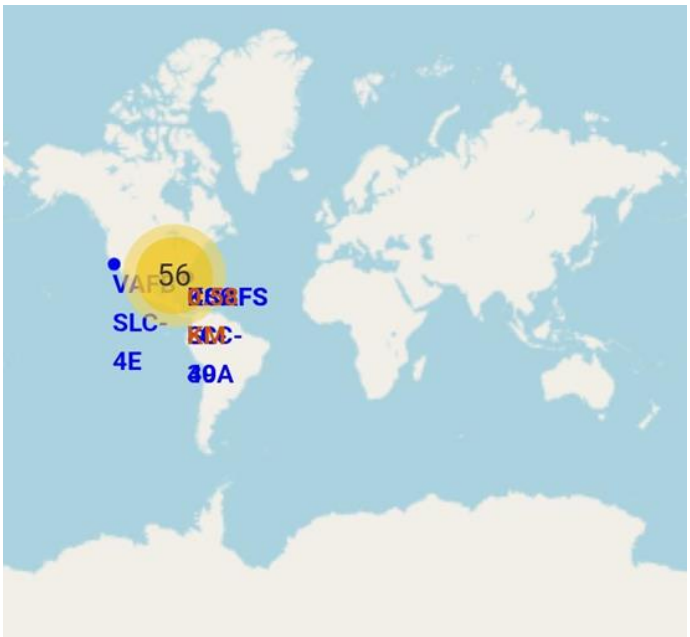
Ocean landings: 5

Success Rate: 38% (8/21 landing attempts).

Section 3

Launch Sites Proximities Analysis

Geographical Distribution of SpaceX Launch Sites.



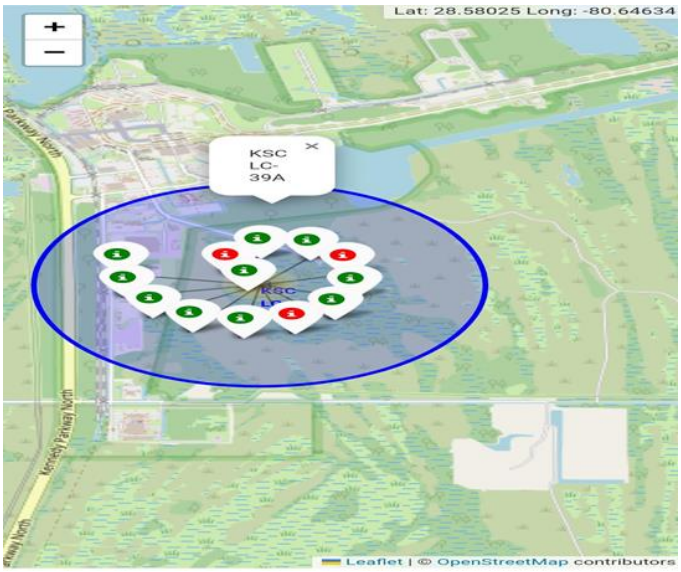
The map shows a global view with launch sites mainly concentrated in the United States, including “CCAFS SLC-40” and “KSC LC-39A,” as well as a cluster marker (“56”) indicating a high density of launches.

Visualizing SpaceX Launch Success Rates by Geographic Location.

Key Elements & Findings:

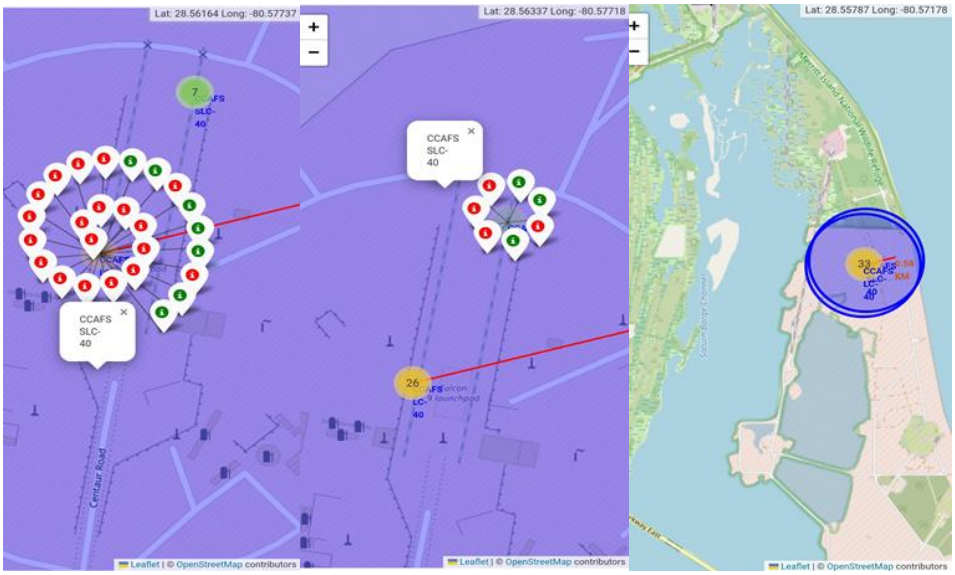
VAFB SLC-4E (34.63°N, -120.61°W) shows 10 launch markers: 4 green (success) and 6 red (failures). The pad label and nearby Dix Road are visible. With a 40% success rate, this site performed worse than Florida sites, likely reflecting more experimental or challenging polar-orbit missions.

Visualizing SpaceX Launch Success Rates by Geographic Location.



Kennedy Space Center (KSC LC-39A): 13 launches with 10 successes and 3 failures (~77% success rate). This makes KSC LC-39A the most reliable site in the dataset—more than double the success rate of CCAFS (~30%). The map shows predominantly green markers within the blue boundary, confirming high operational reliability.

Visualizing SpaceX Launch Success Rates by Geographic Location.



Cape Canaveral Space Force Station (CCAFS):

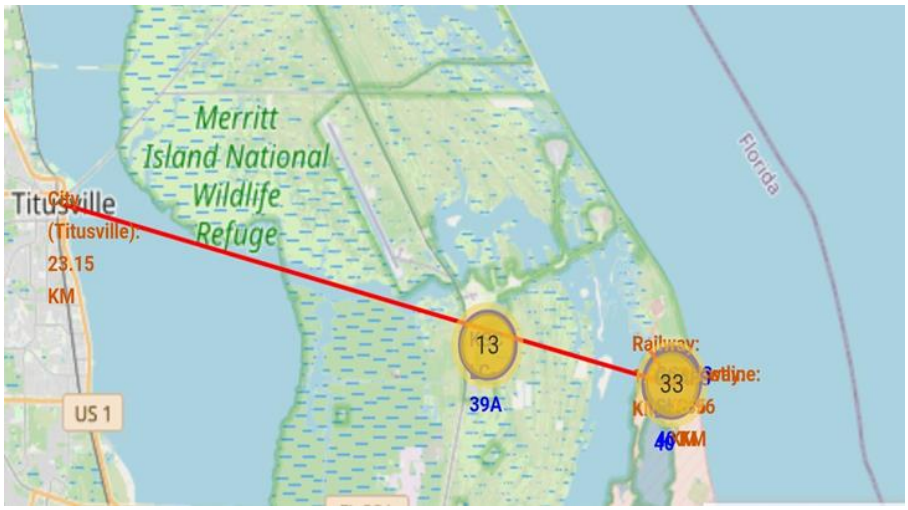
The dataset splits CCAFS into two entries (LC-40 and SLC-40):

Entry 1: 26 launches — 7 success, 19 failures (27%)

Entry 2: 7 launches — 3 success, 4 failures (43%)

Total: 33 launches with about 30% success, much lower than Kennedy Space Center (KSC) (~77%).

Visual Analysis of SpaceX Launch Sites and Landing Outcomes.



CCAFS LC-40 (Cape Canaveral, FL)

Nearest City: Titusville (approx. 25 km)

Logistics: Immediate highway access for equipment; adjacent to the Atlantic coastline (< 1.0 km) for safe over-water trajectories.

KSC LC-39A (Kennedy Space Center, FL)

Nearest City: Titusville (approx. 22 km)

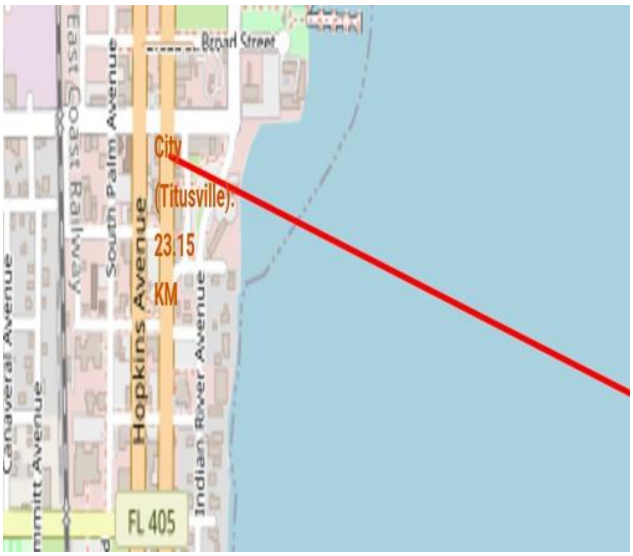
Logistics: Directly served by the NASA Railroad for booster transport; maintains a 1.5 km buffer from the coastline.

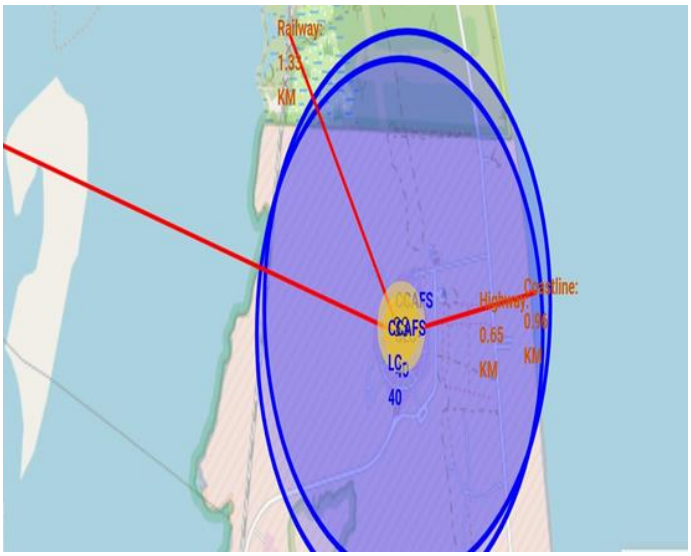
VAFB SLC-4E (Vandenberg AFB, CA)

Nearest City: Lompoc (approx. 15 km)

Logistics: Strategic access to the Coastal Rail Line; positioned < 1.0 km from the Pacific Ocean, ideal for polar orbit launches.

Visual Analysis of SpaceX Launch Sites and Landing Outcomes.

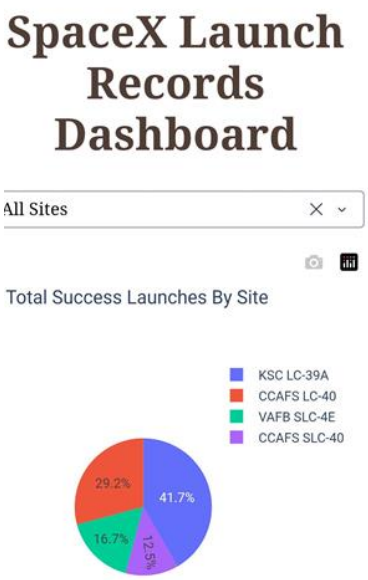




Section 4

Build a Dashboard with Plotly Dash

SpaceX Launch Success Distribution by Site.

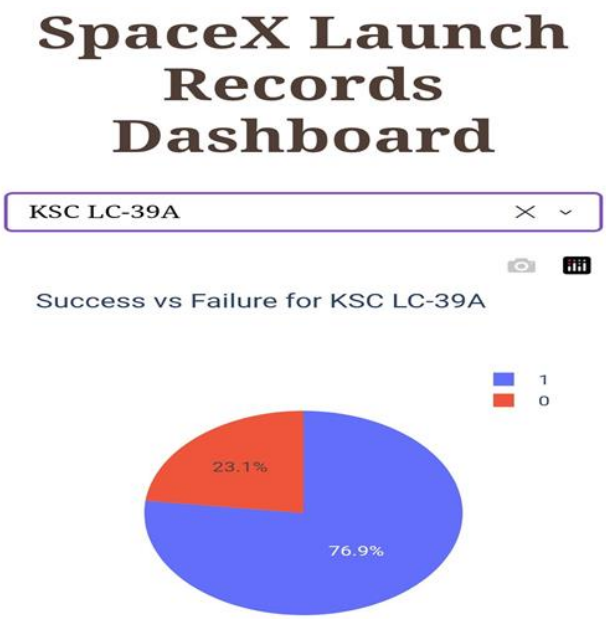


The dashboard titled “SpaceX Launch Records” from SpaceX shows the distribution of successful launches by site using a pie chart, with the filter set to “All Sites.”

Four launch sites are included: KSC LC-39A, CCAFS LC-40, VAFB SLC-4E, and CCAFS SLC-40. The chart highlights that KSC LC-39A leads with 41.7% of successful launches, making it the primary site. CCAFS LC-40 follows with 29.2%, while VAFB SLC-4E accounts for 16.7%, and CCAFS SLC-40 has the smallest share at 12.5%.

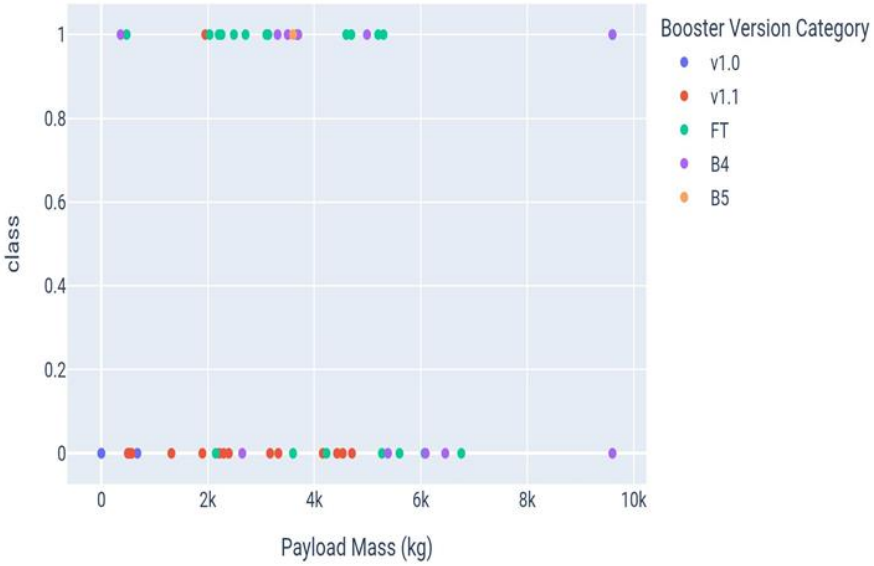
Overall, most launches come from Florida sites, showing a strong concentration of operations there, with KSC LC-39A as the main hub.

KSC LC-39A: Highest Launch Success Rate.



The KSC LC-39A launch site has a 77% success rate and a 23% failure rate, making it one of SpaceX's most reliable locations. The dashboard displays this with a clear pie chart (blue for success, red for failure) and includes an interactive dropdown to select different launch sites. Overall, KSC LC-39A demonstrates consistent and dependable launch performance.

Payload Mass vs. Launch Success: Key Insights.



- As Falcon 9 evolved, newer versions (FT, Block 4/5) started handling much heavier payloads (>6,000 kg), while early versions were limited to lighter ones.
- The most reliable range is 6,000–10,000 kg, where launches had a 100% success rate.
- Even heavier missions (>10,000 kg) are also highly reliable, but only possible with the latest versions.
- Most failures happened with lighter payloads (0–6,000 kg), mainly during early, experimental stages.

Section 5

Predictive Analysis (Classification)

Classification Accuracy.

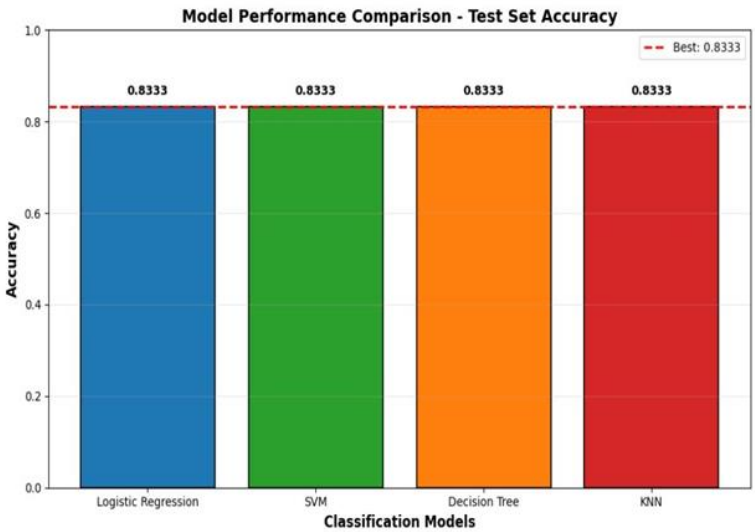


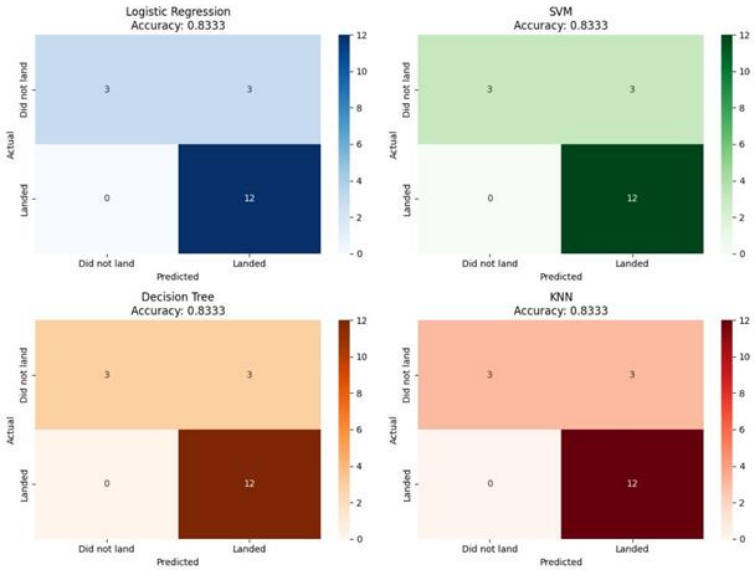
Chart Description:

The bar chart compares the test set accuracy of four classification models: Logistic Regression, SVM, Decision Tree, and KNN.

Highest Accuracy:

All models are tied. Each model achieved an identical accuracy of **0.8333 (83.33%)**.

Confusion Matrix.



Confusion Matrix (Best Model):

TN = 3, FP = 3, FN = 0, TP = 12

Explanation:

Accuracy = 83.33% (15/18 correct). The model detects all landings (no FN) but makes some false positives (predicts landing when it doesn't).

Conclusions.

Falcon 9 landing success improved over time — clear learning effect observed.

Launch site, orbit, and payload strongly impact landing success.

Best model (Decision Tree) achieved ~83–94% accuracy in predictions.

Data-driven insights can help reduce launch costs and operational risks.

Reusability is a critical factor in SpaceX’s competitive advantage.

Appendix.

1. Data sources: SpaceX API and Wikipedia scraping
2. Key code snippets included (API requests, data cleaning, feature creation)
3. SQL queries used for filtering and basic analysis
4. Visual outputs: EDA charts and success trends
5. ML workflow: preprocessing, training, tuning and evaluation
6. Tools used: Python (Pandas, Matplotlib, Folium, Dash)
7. Models tested: Logistic Regression, SVM, KNN, Decision Tree
8. Full notebook available (Colab / GitHub)

Thank you!